

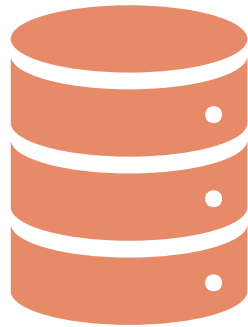
State Managment

Bloc pattern

What is state?



Definition& importance



State is the data that reflects the current status of your application. It can include information such as user inputs, results from an API, or the status of user interaction ,Loading, Success ,Failure



Importance of State: Proper state management ensures that the application responds correctly to data changes, which improves user experience

How does state changes?

User Interaction: For example, clicking a button may change the button text or trigger loading of new data.



Network Response: Receiving new data from an API can lead to changes in the application's state.

Stateful & stateless widgets

How stateless widget rebuild?



Make a children of it in stateful widget

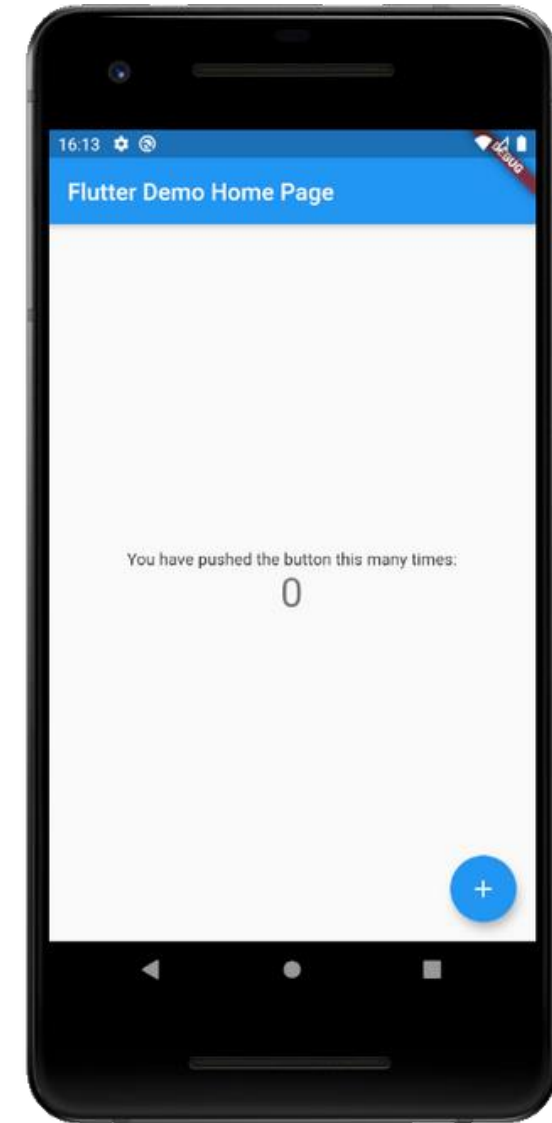


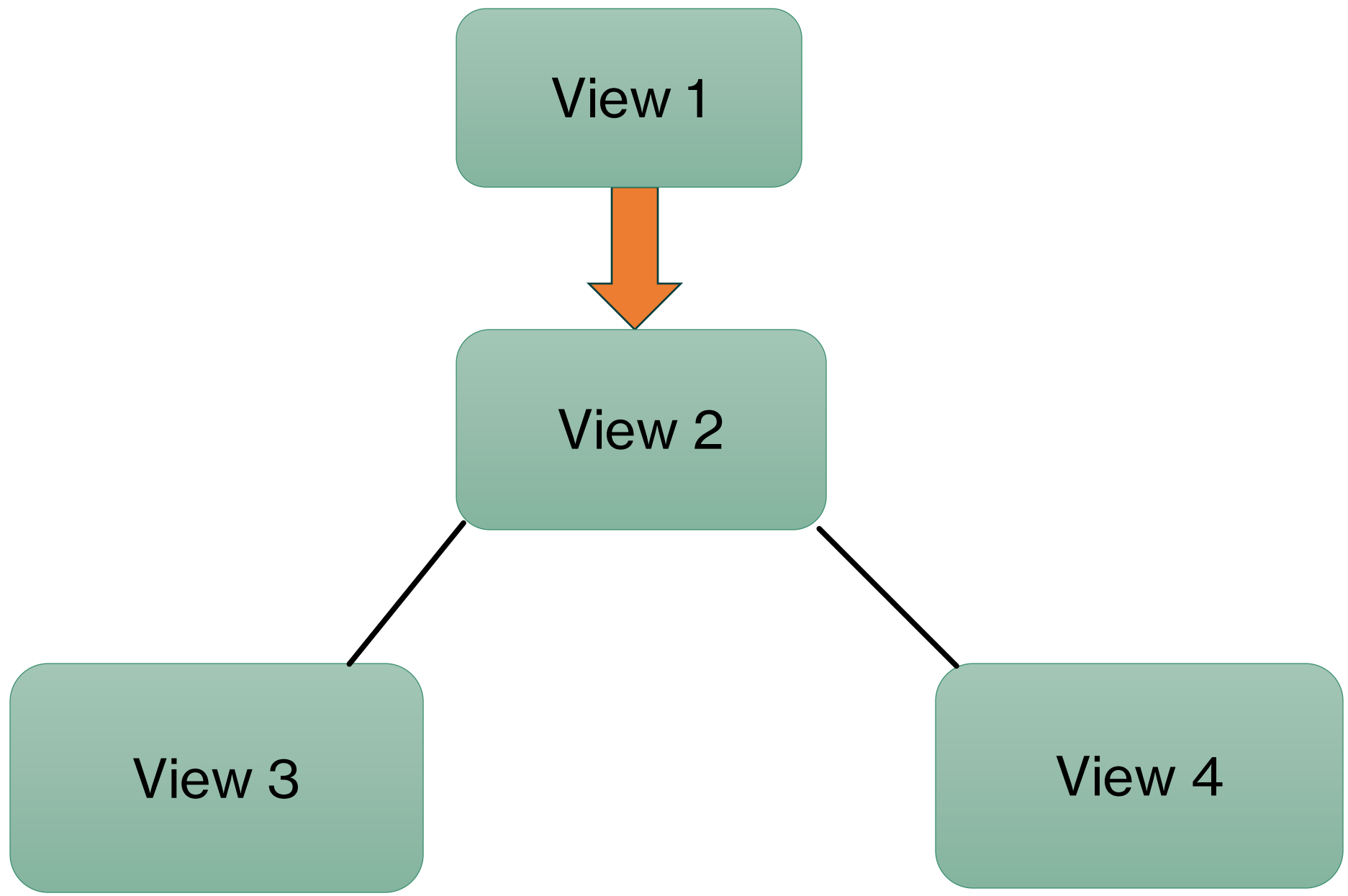
Use State Management

Why state management?

- change UI without **Stateful** widget
- Change only the part that changes such as (text , button,..)

Transfer data between views





State Management approaches

setState

Bloc

Provider

Getx

Riverpod

BLoC Pattern

- **What is the BLoC Pattern?**
- BLoC (Business Logic Component) is a **design pattern** that separates business logic from the user interface. It uses streams to pass data from the BLoC to the UI, and it receives events from the UI.



Benefits of BLoC



Separation of Concerns: It makes it easier to test business logic independently from the user interface.



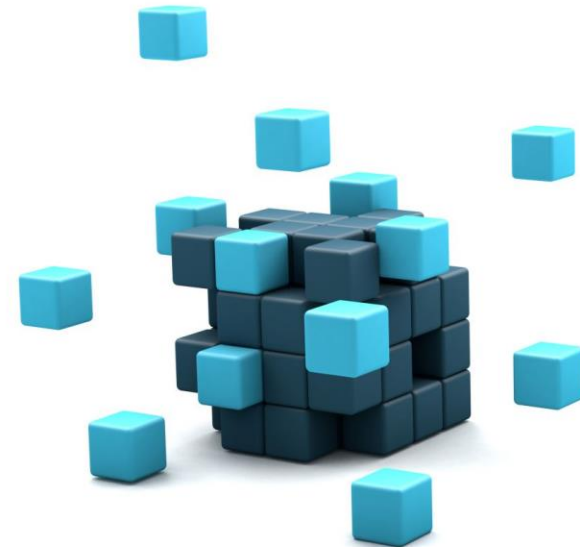
Smooth Interaction: It allows for a more seamless user experience by managing events asynchronously.



Reusability: BLoC components can be reused in different applications with minimal changes.

What is cubit ?

- **Cubit** is a lightweight state management solution that is part of the BLoC (Business Logic Component) package. It's designed to make it easier to handle simple state management needs without the additional complexity of BLoC events and streams. Cubit is particularly useful for applications that require reactive state management but don't need the full structure of the BLoC pattern.



Cubit & Bloc

